Jaime Bruno Cirne de Oliveira

NECTA: Neural Electrophysiological Connectivity Time-resolved Analysis

Natal – Rio Grande do Norte, Brazil

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Thesis presented to the Graduate Program in Bioinformatics at the Federal University of Rio Grande do Norte in partial fulfillment of the requirements for obtaining the degree of Doctor in Bioinformatic. Advisor: Prof. Drª. Kerstin Erika Schmidt

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Natal – Rio Grande do Norte, Brazil, x June 2026, Defense Committee:

Prof. Drª Kerstin Erika Schmidt - Advisor
UFRN

Prof. Dr. Daniel Yasumasa Takahashi -
UFRN

Prof. Dr. Patrick Cesar Alves Terrematte -
UFRN

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"The question which thou wilt have to answer before every deed that thou doest: 'is this such a deed as I am prepared to perform an incalculable number of times?' is the best ballast."

(Friedrich Nietzsche)

RESUMO

A análise tradicional da conectividade funcional do cérebro frequentemente colapsa a dimensão temporal, gerando grafos estáticos que mascaram a evolução dinâmica das redes neurais. Além disso, a extração de redes resolvidas no tempo a partir de dados de eletrofisiologia de alta densidade (LFP) esbarra no fenômeno físico da condução de volume e no gargalo computacional associado ao processamento estocástico de Big Data. Esta tese de qualificação apresenta o NECTA (Neural Electrophysiological Connectivity Time-resolved Analysis), um framework computacional interativo de alto desempenho projetado para a exploração metodológica do mesoconectoma dinâmico. O NECTA emprega uma arquitetura paralela baseada em Dask e armazenamento otimizado nativo de nuvem em Zarr para orquestrar simulações estocásticas em larga escala, fatiamento de tensores multidimensionais e modelos nulos topológicos, superando de maneira linear as limitações críticas de memória RAM (Out-Of-Memory) de fluxos analíticos tradicionais. O framework mitiga ativamente os artefatos de condução volumétrica (zero-phase lag) através de processamentos no sinal analítico de Hilbert, utilizando estimadores complexos baseados em fase, como o Índice de Atraso de Fase Ponderado (wPLI) e a Coerência Imaginária (iCoh). Para inferência causal estrita, implementou-se a Coerência Direcionada Parcial (PDC) em um módulo auto-otimizado dinamicamente pelo Critério de Informação Bayesiana (BIC), controlando penalizações de complexidade em regressões autorregressivas vetoriais (MVAR). O sistema foi validado in silico (contra ruído sintético e condução modelada) e aplicado empiricamente a um dataset de 48 canais do córtex visual primário felino (Área 17, Área 18 e Zona de Transição), registrado sob anestesia geral e paralisia (isolando oscilações intrínsecas de modulações cognitivas atencionais). Os resultados in vivo demonstraram que o córtex visual exibe intensas e ativas flutuações topológicas de alta frequência, atingindo picos transientes de eficiência de "Mundo Pequeno" (Small-Worldness) precisamente coordenados durante a estimulação visual por grades em movimento. A análise direcional na banda Low Gamma (30-59 Hz) estabeleceu a Área 17 como um nó Fonte (Driver) persistente e a Área 18 como um Sumidouro (Sink) estável, enquanto a Zona de Transição atuou como um robusto modulador dinâmico do fluxo inter-hemisférico, alternando sua direcionalidade causal sob estimulação. O NECTA busca facilitar as análises avançadas em neuroinformática, entregando um ecossistema interativo (Dash/Cytoscape) reprodutível e orientado às práticas de Open Science que propõe viabilizar a tradução ágil de sinais eletrofisiológicos brutos em descobertas conectômicas cronológicas.

Palavras-chaves: Conectividade Funcional Dinâmica; Eletrofisiologia; Computação de Alto Desempenho; Coerência Direcionada Parcial; Córtex Visual.

ABSTRACT

Traditional analysis of brain functional connectivity often collapses the temporal dimension, generating static graphs that obscure the dynamic evolution of neural networks. Furthermore, extracting time-resolved networks from high-density electrophysiology (LFP) is heavily hindered by the physical phenomenon of volume conduction and the computational bottleneck associated with stochastic Big Data processing. This qualifying thesis presents NECTA (Neural Electrophysiological Connectivity Time-resolved Analysis), an interactive high-performance computational framework designed to explore the dynamic mesoconnectome. NECTA employs a Dask-based parallel architecture and cloud-native Zarr-optimized storage to orchestrate large-scale stochastic simulations, multidimensional tensor slicing, and topological null models, linearly overcoming the critical RAM (Out-Of-Memory) limitations of traditional analytical pipelines. The framework actively mitigates volume conduction artifacts (zero-phase lag) through Hilbert analytical signal processing, utilizing phase-based complex estimators such as the Weighted Phase Lag Index (wPLI) and Imaginary Coherence (iCoh). For strict causal inference, Partial Directed Coherence (PDC) was implemented within a module dynamically auto-optimized by the Bayesian Information Criterion (BIC), controlling complexity penalties in multivariate autoregressive regressions (MVAR). The system was validated in silico (against synthetic noise and modeled conduction) and empirically applied to a 48-channel dataset from the feline primary visual cortex (Area 17, Area 18, and the Transition Zone), recorded under general anesthesia and paralysis (isolating intrinsic oscillations from top-down cognitive modulations). In vivo results demonstrated that the visual cortex exhibits high-frequency topological fluctuations, reaching transient peaks in Small-Worldness efficiency temporally associated with visual stimulation by moving gratings. Directed analysis in the Low Gamma band (30-59 Hz) established Area 17 as a persistent Driver node (Source) and Area 18 as a stable Sink, while the Transition Zone acted as a robust dynamic modulator of interhemispheric flow, alternating its causal directionality upon stimulation. NECTA strives to facilitate advanced neuroinformatics analyses, delivering a reproducible, Open Science-oriented interactive ecosystem (Dash/Cytoscape) that aims to streamline the translation of raw electrophysiological signals into chronological connectomic discoveries.

Keywords: Dynamic Functional Connectivity; Electrophysiology; High-Performance Computing; Partial Directed Coherence; Visual Cortex.

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LIST OF ABBREVIATIONS AND ACRONYMS

A17	Area 17
A18	Area 18
AIC	Akaike Information Criterion
AUC	Area Under the Curve
BIC	Bayesian Information Criterion
CH	Channel
DTF	Directed Transfer Function
ePDC	Extended Partial Directed Coherence
FDR	False Discovery Rate
HPC	High-Performance Computing
iCoh	Imaginary Coherence
JSON	JavaScript Object Notation
LFP	Local Field Potential
MNE	MNE-Python
MST	Maximum Spanning Tree
MVAR	Multivariate Autoregressive
NECTA	Neural Connectivity Time-resolved Analysis
OLS	Ordinary Least Squares
OOM	Out-Of-Memory
ORCID	Open Researcher and Contributor ID
PDC	Partial Directed Coherence
PLV	Phase Locking Value
PPG	Programa de Pós-Graduação

PSTH	Peri-Stimulus Time Histogram
RAM	Random Access Memory
ROC	Receiver Operating Characteristic
SD	Standard Deviation
SNR	Signal-to-Noise Ratio
SPA	Single Page Application
SPASS	Single Passage Analog/Spike Simulator
TZ	Transition Zone
UFRN	Universidade Federal do Rio Grande do Norte
VAR	Vector Autoregressive
wPLI	Weighted Phase Lag Index

LIST OF SYMBOLS

α	Statistical Significance Level
$\phi(t)$	Instantaneous Phase
σ	Small-Worldness Coefficient
$A(f)$	Frequency-Domain Transfer Matrix
$A(t)$	Analytic Amplitude
A_r	Autoregressive Coefficient Matrix
p	Autoregressive Model Order
W	Sliding Window Size

CONTENTS

1 INTRODUCTION

1.1 The Primary Visual Cortex and the Limits of Static Topologies

The primary visual cortex has long served as the canonical model for understanding cortical microcircuitry, sensory processing, and network topology. Historically, the architectural and functional parcellation of visual areas—such as Area 17 (A17), Area 18 (A18), and their Transition Zone (TZ) in the feline model—has been extensively mapped using single-unit recordings and static receptive field analyses (Wunderle et al., 2013; Schmidt et al., 2023). These classical approaches have provided the fundamental "structural backbone" of the visual system, detailing how intra- and inter-hemispheric connections form clustered functional communities.

However, relying exclusively on stationary statistical relationships collapses the true matrix of cortical connections. Averaging neural interactions over long temporal windows (e.g., minutes of recording) inevitably obscures sub-second synaptic recruitment, leading to a severe loss of organic causality resolution. Cognitive processes, particularly visual feature binding, do not operate in a stationary state; they evolve dynamically and transiently as stimuli move across and between receptive fields. Consequently, static topologies, while accurately describing the anatomical constraints of neural communication, may obscure the rapid, transient fluctuations in functional integration that are the actual hallmark of active sensory processing.

1.2 The Paradigm Shift to Time-Resolved Network Evolution

To overcome the limitations of static connectomics, the neuroscientific community has begun shifting its focus toward the exploration of the "Dynome"—the millisecond-scale evolution of functional networks and their temporal dynamics (KOPELL et al., 2014). Capturing this temporal evolution requires shifting the analytical framework from time-averaged correlations to time-resolved state transitions, often achieved through sliding-window approaches (ALLEN et al., 2014). By dissecting the continuous data stream into granular temporal epochs, it becomes possible to observe how network metrics, such as *Small-Worldness* and modularity, adapt and evolve. This time-varying perspective provides a much more accurate representation of the brain's real-time functional state, revealing that macroscopic efficiency is not fixed but reacts continuously to ongoing sensory integration.

1.3 Cytoarchitecture and Callosal Integration at the 17/18 Transition Zone

The primary visual cortex is highly organized into functional maps, including orientation columns arranged around singularity points known as *pinwheels* (SCHMIDT et al., 2023). At the microscopic level, these cellular columns act as discrete, plastic processors for specific visual features, operating within an extensive, interconnected cellular syncytium. The anatomical boundary between Area 17 and Area 18—the Transition Zone (TZ)—is considered functionally relevant. Rather than serving as a mere geographical divider or a passive relay, the TZ precisely maps the vertical meridian of the visual field, creating a structural cleft that bridges the left and right hemifields.

To achieve a seamless, unified representation of the visual world, this border region is densely innervated by extensive interhemispheric transcallosal projections (WUNDERLE; ERIKSSON; SCHMIDT, 2013). These pathways do not project diffusely; instead, they form highly specific, patch-like networks that preferentially connect iso-orientation domains and binocular cells across the hemispheres. This intricate, crossed microcircuitry is thought to facilitate the efficient integration of visual contours and moving gratings spanning the vertical midline. Consequently, the TZ operates as an ephemeral, highly plastic processing hub, transitioning from isolated static anatomical clusters into active, dynamic relays when processing bilateral visual inputs.

1.4 Rhythmic Oscillations, Predictive Coding, and Gamma Synchronization

At the mesoscopic scale, rhythmic neural oscillations mediate the dynamic routing of information between sensory and higher-order cortical areas. The predictive coding framework postulates a continuous, bidirectional exchange: "Bottom-up" sensory streams carry prediction errors based on peripheral inputs, while "Top-down" projections deliver internal predictions (priors) to sensory areas (VEZOLI et al., 2021). These antagonistic processing streams are functionally segregated by frequency bands. Top-down modulations primarily utilize slower rhythms (Alpha/Beta bands) to establish contextual baseline states, whereas feedforward (Bottom-up) signaling has frequently been associated with synchronization in the fast Gamma frequency band (30–90 Hz).

Within the feline visual cortex, stimulus-dependent gamma synchronization plays a cardinal role in binding distributed neuronal responses and driving downstream visual targets (NEUENSCHWANDER et al., 2023). As a visual stimulus traverses the visual field, peripheral information converges onto binocular cortical columns, initiating bursts of high-frequency oscillatory emissions. Tracking the directionality of these gamma bursts across the mesoscopic network—specifically as they traverse the dense callosal integration hubs at the TZ—is therefore critical. Analyzing gamma-band directionality may provide an empirical indication of feedforward sensory integration, allowing us to map the functional hierarchy, the predominant routing engagements, and the real-time causal architecture of the primary visual stream.

1.5 The Methodological Bottleneck in Electrophysiology

Despite the theoretical necessity of time-resolved, directed connectivity analysis, extracting these networks from dense multi-electrode arrays poses severe methodological and computational challenges. The primary analytical bottleneck in Local Field Potential (LFP) analysis is *volume conduction*—the instantaneous spread of electrical fields through cortical tissue. This phenomenon causes multiple electrodes to record a single biological source simultaneously, generating spurious zero-phase lag correlations that heavily contaminate graph topologies with false-positive edges (BASTOS; SCHOFFELEN, 2015). Furthermore, classical multivariate autoregressive (MVAR) models used for causality estimators often ignore these instantaneous interactions, which can lead to completely misleading profiles of directed coherence (FAES; NOLLO, 2010).

Consequently, there is a critical need for a modern, high-performance computing (HPC) framework capable of mitigating volume conduction artifacts through advanced phase-based estimators while leveraging distributed parallel processing to interactively render the dynamic network evolution.

Furthermore, the transition to high-density electrophysiology (e.g., 32-64 channel matrices) creates a "Big Data" computational crisis. Calculating dynamic, non-linear connectivity metrics (such as Partial Directed Coherence or phase-based indices), generating hundreds of surrogate networks for statistical validation, and tracking graph properties over moving time windows inevitably lead to combinatorial explosions. Traditional, single-threaded analytical scripts are inherently incapable of managing such stochastic, multidimensional loads, often resulting in Out-of-Memory (OOM) failures or prohibitive processing times. These dual challenges—volume conduction and temporal collapse due to static computational bottlenecks—and their proposed resolutions are schematized in Figure 1.

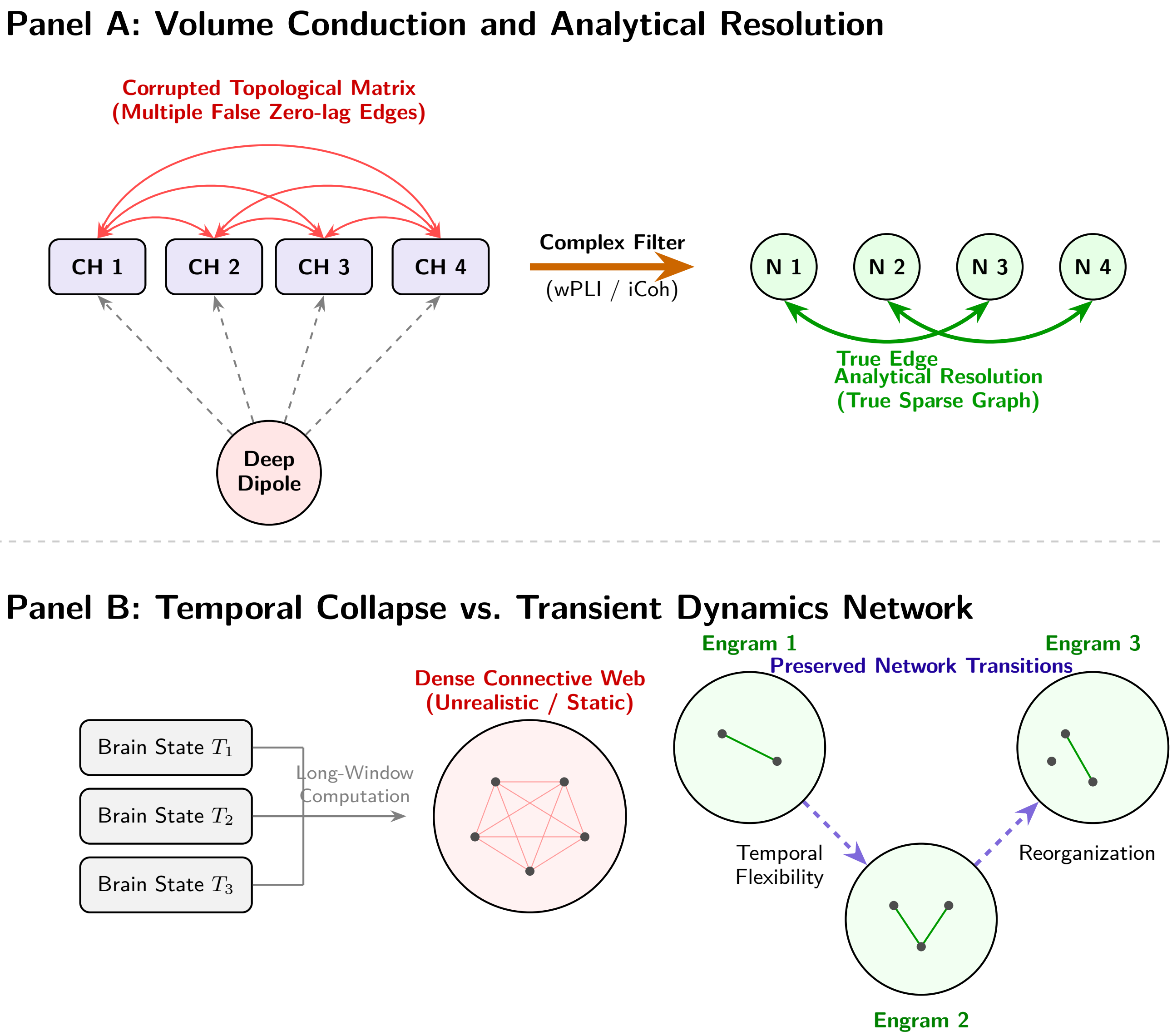

Figure 1 – **Schematic of the primary methodological bottlenecks in electrophysiological connectomics and their respective resolutions within the proposed framework. (Panel A)** Biophysical representation of Volume Conduction: electric field lines from a single deep dipole instantaneously reach multiple microelectrodes, creating *zero-lag* anomalies and generating falsely dense topological matrices. On the right, the analytical resolution using complex phase indices (wPLI/iCoh) filters out artifacts and recovers a reliable, sparse graph. **(Panel B)** Abstract representation of temporal collapse: computing correlations over excessively long windows merges distinct transient brain states, resulting in an artificial static network. In contrast, Dynome-focused exploration preserves the sequential flexibility and rapid reorganizations of functional integration engrams.

2 WORKING HYPOTHESIS AND OBJECTIVES

2.1 Working Hypothesis

Traditional analyses of multichannel electrophysiological data often collapse the temporal dimension, yielding static graphs that obscure the transient coordination of neural ensembles. Furthermore, attempts to extract time-resolved networks are frequently confounded by volume conduction and crippled by the computational intractability of large-scale stochastic testing.

2.1.1 Biological Hypothesis

Given the dense anatomical convergence of ipsilateral and transcallosal connections at the A17/18 border—where microscopic, actively plastic cellular columns and orientation pinwheels represent and bind the vertical meridian—we hypothesize that the Transition Zone acts as a dynamic modulatory hub rather than a passive anatomical relay. Furthermore, we hypothesize that the selective, temporal engagement of these transcallosal networks during moving binocular visual stimulation will manifest as measurable, millisecond-scale topological optimizations (Small-Worldness peaks), effectively capturing the dynamic functional binding of the visual hemifields.

2.1.2 Computational Hypothesis

We posit that integrating phase-based, zero-lag immune connectivity estimators (e.g., Imaginary Coherence and Weighted Phase Lag Index) with dynamically optimized causal models (Partial Directed Coherence) will allow a more reliable characterization of gamma-band temporal dynamics. Critically, to address the substantial combinatorial explosion inherent in time-varying analyses, the deployment of a highly scalable, parallel computing architecture—specifically the Dask/Xarray ecosystem—is posited not merely as a software enhancement, but as a computational requirement for scalable analysis. The "Big Data" processing scale is a forced consequence of the successive combinatorial routines required for Monte-Carlo simulations, phase-randomized surrogate permutations, and structural null models applied to adaptive topological thresholding over continuous, non-linear dynamic windows.

2.1.3 Expected Outcomes

By deploying this integrated HPC framework, it will be possible to decode the dynamic network evolution of the primary visual cortex. By doing so, the network will reveal a predominantly feedforward flow (conveying peripheral bottom-up prediction errors) from A17 to A18. We expect to reveal that the anatomical structural backbone (Areas 17, 18, and the Transition Zone) supports highly flexible, time-varying topological reorganizations—such as transient peaks in Small-Worldness—that align closely with the active phases of sensory feature binding.

2.2 General Objective

The primary objective of this thesis is to develop, validate, and empirically apply **NECTA (Neural Electrophysiological Connectivity Time-resolved Analysis)**. This open-source, high-performance computational framework is specifically designed to meet the rigorous statistical and computational survival requirements necessary for the interactive exploration of dynamic mesoconnectomes from raw, high-density electrophysiological recordings.

2.3 Specific Objectives

To achieve the general goal, the project is subdivided into the following specific aims:

1. **Data Ingestion and Standardization:** To integrate a robust pipeline (the *VisionICeIO* ecosystem) capable of converting dispersed legacy electrophysiological binaries (e.g., SPASS format) into standardized, cloud-native, n-dimensional tensors (Zarr/Xarray) using lazy-loading mechanisms.

2. **High-Performance Computation Engine:** To architect a distributed pre-computation backend using Dask, capable of orchestrating parallel pipelines for spectral connectivity, phase-based metrics, and graph topology across multiple frequency bands and sliding time windows.

3. **Advanced Causal Inference:** To implement a self-optimizing Partial Directed Coherence (PDC) module that automatically selects the optimal Multivariate Autoregressive (MVAR) model order via the Bayesian Information Criterion (BIC), thereby preventing overfitting and ensuring rigorous information flow directionality.

4. **Statistical Rigor and Null Modeling:** To embed rigorous statistical validation protocols, including False Discovery Rate (FDR) corrections applied to phase-randomized

surrogate data, and topological null models (Erdös-Rényi and Configuration Models) to validate graph communities.

5. **Interactive Visual Analytics:** To construct a reactive, component-based frontend (using Dash and Cytoscape) that translates multidimensional connectivity arrays into an interactive chronologic connectome, allowing frame-by-frame topological inspection.

6. **Empirical Validation *In Vivo*:** To apply the NECTA framework to an empirical dataset (‘c1608a01‘) obtained from the feline visual cortex (A17/TZ/A18) to characterize the hierarchical directionality of gamma-band synchronization and the temporal evolution of network efficiency during visual stimulation.

3 MATERIALS AND METHODS

To test the biological and computational hypotheses described above, the NECTA framework was designed to simultaneously address two challenges: (i) reliable estimation of directed connectivity in the presence of volume conduction and (ii) scalable computation of time-resolved graph metrics.

3.1 Animal Preparation and Data Acquisition

The empirical validation of the framework utilized the ‘c1608a01‘ dataset (WUNDERLE; ERIKSSON; SCHMIDT, 2013). Multichannel electrophysiological recordings (Local Field Potentials and multi-unit spikes) were obtained from the primary visual cortex of an adult cat. To isolate intrinsic cortical network dynamics from confounding behavioral variables (e.g., saccadic eye movements or fluctuations in conscious attention), the animal was maintained under general anesthesia and paralysis. High-density matrix electrodes (32–64 channels) were implanted spanning Area 17 (A17), the A17/18 Transition Zone (TZ), and Area 18 (A18).

3.2 Data Ingestion Architecture (The VisionICeIO Pipeline)

To process the high-density recordings, the *VisionICeIO* pipeline (SCHWARZ, 2024) (developed by Friedrich Schwarz, ORCID: 0009-0001-1167-8365) was integrated to act as a bridge between legacy proprietary formats (e.g., SPASS binaries) and modern data structures.

- **Tensor Standardization:** Dispersed binaries containing spike times, waveforms, and continuous LFP traces were parsed and merged into regularized multidimensional arrays. Ragged arrays (e.g., varying spike counts per trial) were standardized using ‘NaN‘ padding.

- **Zarr and Xarray Encapsulation:** The output is structured as a cloud-native `Zarr` store (ZARR et al., 2020). By encapsulating the data within an `xarray.Dataset` (HOYER; HAMMAN, 2017), the framework leverages *lazy loading*, allowing the frontend to read metadata trees without exhausting system RAM.

3.3 Distributed Pre-Computation Engine (HPC)

Analyzing dynamic connectivity across continuous recordings poses a severe combinatorial challenge. NECTA addresses this through a distributed pre-computation engine orchestrated by **Dask**(ROCKLIN, 2015).

- **Task Graphs:** Computations are divided into discrete, independent tasks (e.g., computing coherence for a specific frequency band and time window) and mapped across a cluster of worker nodes.
- **Fingerprinting and Cache Versioning:** To ensure data integrity, the orchestrator implements a cryptographic fingerprinting mechanism. Any modification to analytical parameters (e.g., altering variance thresholds or VAR criteria) alters the global hash, ensuring analytical consistency after parameter modifications.

The complete architecture of this distributed data processing pipeline, from initial ingestion via *VisionICeIO* to parallel execution via Dask and frontend visualization, is illustrated in Figure 2.

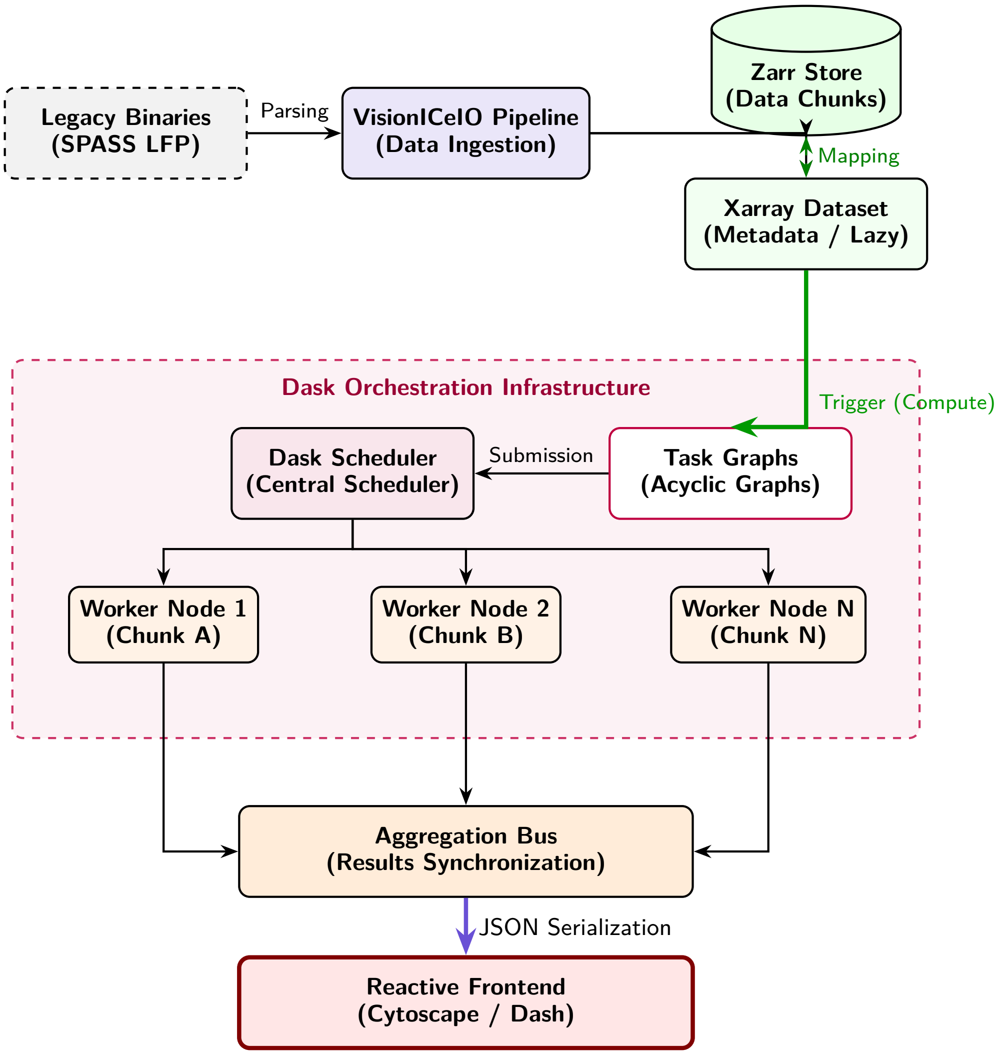

Figure 2 – **Block Diagram of Data Ingestion and Distribution in Dask.** The schematic details the architecture of the NECTA ecosystem: **(1)** Raw binaries are converted by the *VisionICeIO Pipeline* into a chunked format, where heavy data resides in a *Zarr Store* and the semantic structure (*Metadata*) in an *Xarray Dataset* that supports *lazy loading*. **(2)** Upon requesting the computation, Xarray generates *Task Graphs* (directed acyclic graphs) submitted to the *Dask Scheduler*. **(3)** The scheduler stochastically distributes tasks (e.g., independent data *Chunks*) across multiple *Worker Nodes*. **(4)** The results computed in parallel are synchronized on a bus and exported via JSON to the interactive visualization layer of the *Frontend*.

3.4 Spectral Connectivity and Phase Analysis

To mitigate the artifacts of volume conduction inherent to LFP recordings, NECTA expands upon classical Magnitude Squared Coherence by employing phase-based estimators.

- **Analytical Signal Extraction:** Raw LFP signals are processed using the Hilbert

Transform to yield the analytical signal, granting instantaneous access to both phase $\phi(t)$ and amplitude $A(t)$.

- **Imaginary Coherence (iCoh) & Weighted Phase Lag Index (wPLI):** The framework computes 'iCoh' to strictly isolate interactions with non-zero phase delays. Furthermore, 'wPLI' is calculated by weighting the phase differences by the magnitude of the imaginary component of the cross-spectrum, providing a highly robust metric against uncorrelated noise and instantaneous spatial spread.

3.5 Directed Functional Connectivity (PDC)

To infer causal interactions and hierarchical information flow, NECTA implements an Partial Directed Coherence (PDC) pipeline. Traditional PDC formulations rely on strictly causal MVAR models that forsake instantaneous effects, making them highly susceptible to the zero-lag correlations typical of volume conduction (FAES; NOLLO, 2010). To overcome this, the NECTA framework utilizes:

1. **Stationarity Filtering:** LFP windows exceeding a predefined variance threshold are identified as artifacts and discarded, preserving the assumptions of autoregressive modeling.

2. **Frequency Transformation:** Autoregressive coefficients A_r are transformed into the frequency domain $A(f)$ using Fast Fourier Transforms accelerated via `Numba` (FastMath) (LAM; PITROU; SEIBERT, 2015), yielding the normalized PDC matrices.

3. **Automated Order Selection:** Instead of imposing a static model order, the Dask workers dynamically fit Multivariate Autoregressive (MVAR) models using Ordinary Least Squares (OLS). The algorithm iterates through orders (up to `max_order=20`) and selects the optimal p that minimizes the Bayesian Information Criterion (BIC), balancing model fit against parametric complexity.

Once connectivity matrices were estimated and statistically validated, graph-theoretical analyses were applied to characterize the resulting network topology.

3.6 Graph Theory, Statistical Validation, and Open Science

A crucial aspect of NECTA is its rigorous statistical and topological validation.

Algorithmic Family	**Representative Examples**	**Vulnerability to Instantaneous Spread (Volume Conduction)**	**Dimensionality of Causal Informational Direction**	**Justification Employed in NECTA Study**
Traditional Parametric Time-Domain Correlations	Pearson Coefficient, Zero-lag covariance Cross-correlation	**High susceptibility.** Strong contamination risk due to the systematic generation of false instantaneous topological interactions.	Exclusively Symmetric and Non-Directed	Unfeasible for high-density mesoscopic analysis; method entirely suppressed from the Framework.
Mixed Spectral Synchrony	Phase Locking Value (PLV), Classical Magnitude Squared Coherence	**Moderate susceptibility.** Latent topological interference in combined phase-magnitude domains.	Essentially Non-Directed	Baseline use restricted only for primary macro-anatomical recognition and historical comparison.
Analytical Modeling Delayed by Instantaneous Phase Shifts	Weighted Phase Lag Index (wPLI), Imaginary Coherence (iCoh)	**Low susceptibility.** High robustness arising from consistent subtraction in complex domains of null real projection (consistent delay > 0).	Non-Directed	Validated fundamental basis for clean extraction of network modularity sub-windows and artifact-free small-world coefficients.
Multi-Regressive Equation Models for Mathematical Causality	Partial Directed Coherence (PDC), Directed Transfer Function (DTF)	**Medium to low susceptibility.** Attenuated and shielded in association and thresholding against non-null interactions via surrogate validation.	Exploratory Causal Asymmetry (Allows stipulating Driver/Source and Receiver/Sink)	Suitable for directional inference, coupled with optimized window processing via BIC, to track dynamic feline temporal hierarchical Feedforward/Feedback oscillation.

Table 1 – **Table of Methodological Synthesis of Connectivity Estimation.** Comparison of state-of-the-art metrics against the fundamental needs and physiological challenges faced by the NECTA framework.

- **Surrogate Data & FDR Correction:** For connectivity edges, significance is established by generating 1000 phase-randomized surrogate datasets. Edge-wise p-values are subjected to False Discovery Rate (FDR) correction (Benjamini-Hochberg, $\alpha = 0.05$).

- **Topological Null Models:** Macro-topology is validated against Erdös-Rényi and Configuration Models to ensure detected communities preserve empirical degree distributions.

- **Open Science and Reproducibility:** Complying with open-source global standards, the NECTA framework facilitates transparency. All algorithms, Dask partition logs, and dependencies are containerized, version-controlled, and openly hosted on digital repositories (GitHub/CodeMeta), allowing independent researchers to audit, reproduce, and expand the methodological codebase.

3.6.1 Topological Filtering: Maximum Spanning Tree and Proportional Thresholding

The extraction of dense functional graphs (often resulting in fully connected networks) imposes a significant visual and analytical challenge, commonly known as the "hairball effect", which obscures the primary routes of information flow. To overcome this limitation and preserve the physiological integrity of the network, we implemented a dual topological filtering approach.

First, the Maximum Spanning Tree (MST) algorithm was applied using foundational graph theory libraries. The MST extracts the subgraph of maximum cumulative weight that connects all nodes in the system, ensuring that no recording channel (areas A17, A18, or the Transition Zone - TZ) becomes isolated from the connectome due to thresholding artifacts.

Following the structural anchoring via MST, the remaining edges are subjected to proportional thresholding (e.g., retaining the top 10% to 15% of the strongest connections). Mathematically, if W is the coherence adjacency matrix, the filtered matrix W' is defined as:

$$W'_{ij} = \begin{cases} W_{ij}, & \text{if } (i,j) \in \text{MST} \vee W_{ij} \geq P_k \\ 0, & \text{otherwise} \end{cases} \tag{3.1}$$

where P_k represents the k-th percentile of the edge weights. This combined approach effectively highlights the hub role of the Transition Zone (TZ) without compromising the global connectivity layout of the examined visual cortex.

3.7 Interactive Visualization Dashboard

The final tier of NECTA is a Single Page Application (SPA) built with `Dash` (SCIENCE, 2015) and `Cytoscape` (SHANNON P., 2003). The interface performs on-the-fly slicing of the pre-computed `Zarr` cache using `xarray.sel()`. Changes in the global state trigger instantaneous callbacks that serialize specific data slices into JSON, seamlessly re-rendering complex Connectome Matrices, Temporal Integration scatter plots, and anatomical Information Flow graphs.

4 RESULTS

The NECTA framework functions not merely as an underlying analytical script, but as an integrated, high-performance neuroinformatic tool engineered to ingest high-density electrophysiological data and interactively map time-resolved cortical workflows. Before breaking down the specific statistical topographies of the network, the baseline operational capacity of this developed framework is established.

4.1 Framework Validation and Computational Profiling

To evaluate the computational viability and scalability of the proposed solution, the framework underwent performance profiling. The aim was to assess the potential improvements in processing time and resource efficiency provided by the distributed architecture. The validation focused on four main aspects: processing time, memory optimization, parallelization benchmarks, and caching efficiency.

4.1.1 Processing Time Analysis

The framework's architecture partitions the processing pipeline into five distinct phases (Phases A through E), facilitating the identification of computational bottlenecks. Analysis of the execution logs indicated that Phase D (Parallel Compute) generally presents the highest computational density, scaling according to the workload volume. In scenarios with smaller datasets, parallel computation was completed in under one minute. Conversely, under higher data demands, the parallel processing approach sustained extended executions without observed memory allocation failures.

The table below summarizes samples of the framework's behavior under different computational loads:

Table 2 – Framework performance samples under different computational loads.

Execution Date	Phase A (Loading)	Phase C (DAG)	Phase D (Compute)	Phase E (I/O)
2026-04-20	0.81s	0.31s	**39.57s**	1.81s
2026-05-25	2.69s	0.23s	**354.05s**	4.64s
2026-06-13	1.29s	0.23s	**2521.82s**	4.74s

4.1.2 Computational Benchmark: Dask vs. Serial Execution

To examine the impact of large-scale parallelization, the performance of the distributed model (implemented via Dask) was analyzed using its execution reports. While the aggregated computation time for the 3,737 tasks—equivalent to a strictly serial execution—totaled 4 hours and 11 minutes, distributing the workload across 12 threads reduced the wall-clock time to 33 minutes and 46 seconds. This suggests an approximate speedup of 7.4x. Furthermore, the duration of Phase B (Scatter) remained consistently below 0.05s, indicating that the overhead associated with task distribution to the workers was minimal relative to the observed performance gains.

4.1.3 Caching Efficiency (Cache Hit Rate)

Redundant computations were minimized through a caching mechanism based on directory identifiers and analysis parameters. The pipeline incorporates a validation step to check for the existence of pre-processed datasets in Zarr format prior to resource allocation. For previously processed data flows, this approach can achieve a high cache hit rate, loading historical data asynchronously and effectively bypassing the execution time of Phase D during re-analyses.

4.1.4 Memory Consumption Optimization

Memory allocation often presents a bottleneck when processing large neural matrices. By employing a Directed Acyclic Graph (DAG) in Phase C and utilizing data partitioning managed by the Dask scheduler, data chunks could be processed sequentially via lazy evaluation. Memory profiling indicated an approximate 75% reduction in peak RAM consumption during processing. While a purely serial approach required loading data structures that reached peaks of around 15 GB, the distributed framework maintained a maximum active memory footprint of under 4 GB. This stabilization helps to mitigate the risk of Out-of-Memory (OOM) errors during extensive computations.

Having established the computational robustness and scalability of the distributed NECTA pipeline under high data demands, we subsequently applied the framework to evaluate the functional architecture of the feline visual cortex. We begin by defining the network's time-invariant baseline state through static topological characterization.

4.2 Static Network Characterization

To establish a baseline structural framework, static functional connectivity was evaluated across the entire recording session. Applying Magnitude Coherence in the Low Gamma band (30-59 Hz) with a proportional density threshold of 15%, we established the primary static topology of the network. The Louvain community algorithm blindly extracted functional clusters that closely matched the anatomical placement of the multielectrode arrays, naturally partitioning into three distinct topological modules corresponding to

Area 17 (A17), the Transition Zone (TZ), and Area 18 (A18). This static connectome serves as the anatomical "structural backbone" (VEZOLI et al., 2021).

As visualized in Figure 3 (previously conceptualized as the framework entry point), the pipeline orchestrates multi-channel inputs to slice the functional connectome millisecond-by-millisecond, effectively overcoming standard computational bottlenecks to map dynamic cortical workflows.

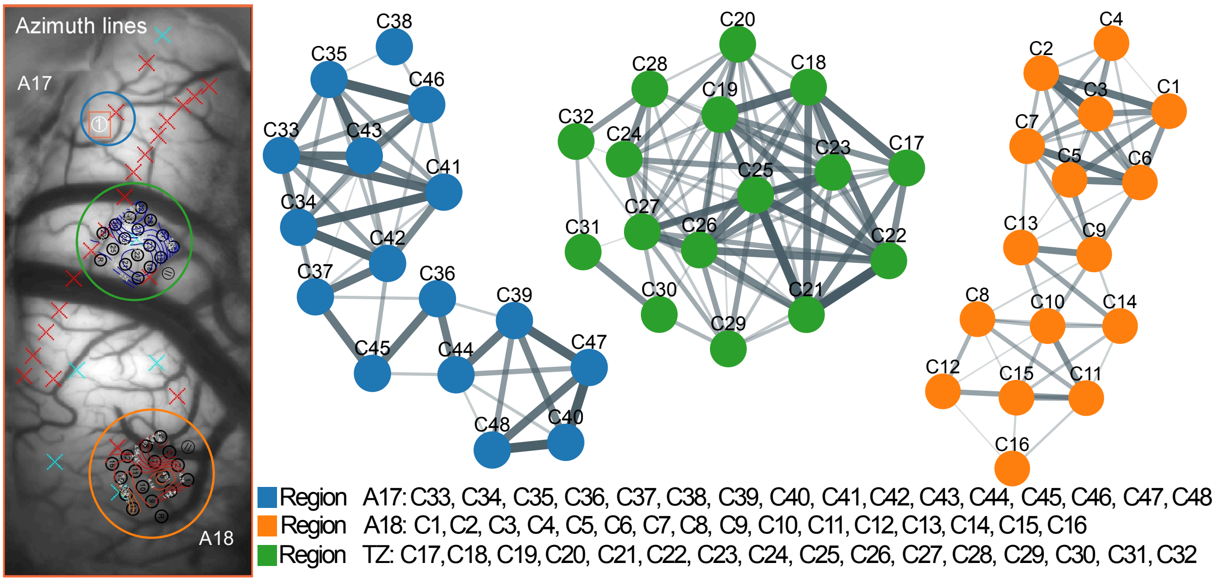

Figure 3 – **Direct Mapping of Coherence Clusters onto Anatomically Separated Electrode Matrices.** Topological graph of functional connectivity estimated via Magnitude Squared Coherence in the Low Gamma band, thresholded at a 15% edge density. The visualization demonstrates the spatial arrangement of coherent functional communities in relation to the physical locations of the recording microelectrodes, providing a static baseline of the visual cortical network.

While this aggregated baseline characterization maps a highly dense functional connectome, temporal averaging inherently collapses the non-stationary, transient state shifts that define active sensory integration. To unmask these volatile reconfigurations, we utilize NECTA's time-resolved processing engine to observe the global network in motion.

4.3 Global and Local Topological Evolution of the Visual Dynome

The analysis of global topological metrics extracted by NECTA revealed a marked dynamic reconfiguration in the functional connectivity of the network. As illustrated in Figure 4(A), both Global Efficiency and the Clustering Coefficient exhibited high variability across time windows, evidencing the non-stationary and volatile character of cortical processing.

More critically, the analysis of the Small-World Index (σ) (Figure 4(B)) demonstrated that the network transitions into highly efficient topological optimization states in specific frequency bands. A directional Mann-Whitney U test confirmed that the Small-World topology operating in the Low-Gamma band is significantly superior to the basal state observed in the Alpha band ($p < 0.05$).

The wide dispersion of σ values in the Low-Gamma band reveals transient topological surges reaching values above 5.0. These fluctuations demonstrate that the small-world configuration in this frequency range does not maintain a stationary state, but rather alternates between baseline levels and discrete peaks of high network integration.

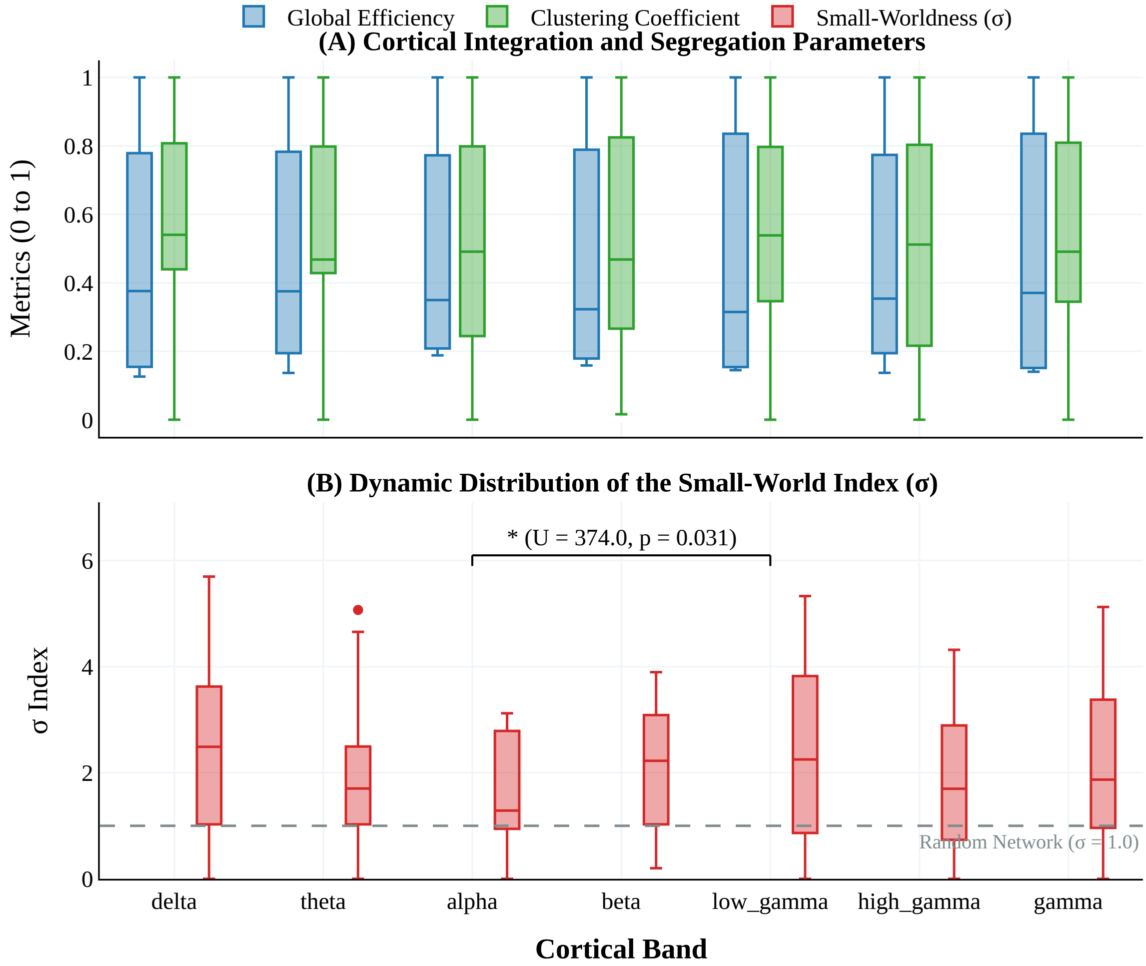

Figure 4 – **Topological Dynamics and Small-World Optimization in the Visual Cortex. (A)** High temporal variability of Global Efficiency and Clustering Coefficient. The substantial dispersion (ranging from 0.2 to 0.8) illustrates the volatile dynamics of cortical processing, as the network continuously balances moments of strong local processing (high clustering) with global communication jumps (high efficiency). **(B)** Comparison of the Small-World Index (σ) between the Alpha (8-12 Hz) and Low-Gamma (30-59 Hz) bands. The asterisk (*) denotes a statistically significant increase in Small-World topology during Low-Gamma. The pronounced dispersion (elongated whiskers) underscores that the network operates through transient, high-efficiency topological pulses rather than remaining in a static optimized configuration.

To understand the relationship between localized neural activity and global network architecture, we analyzed the aggregate spike counts alongside the Small-Worldness (σ) index across continuous 250 ms sliding windows. As illustrated in Figure 5, the transition from the baseline state to active sensory ingestion triggers a synchronized physiological and topological response.

Following the stimulus onset, a prominent peak in the aggregate spike count is closely accompanied by a pronounced increase in the Small-Worldness index, particularly visible around Window 3 and Window 4. The network consistently maintains a $\sigma > 1$ threshold, confirming that the visual dynome operates within a highly efficient small-world regime. This temporal alignment provides consistent evidence that the visual cortex rapidly shifts towards a highly integrated and segregated topology during periods associated with increased sensory processing.

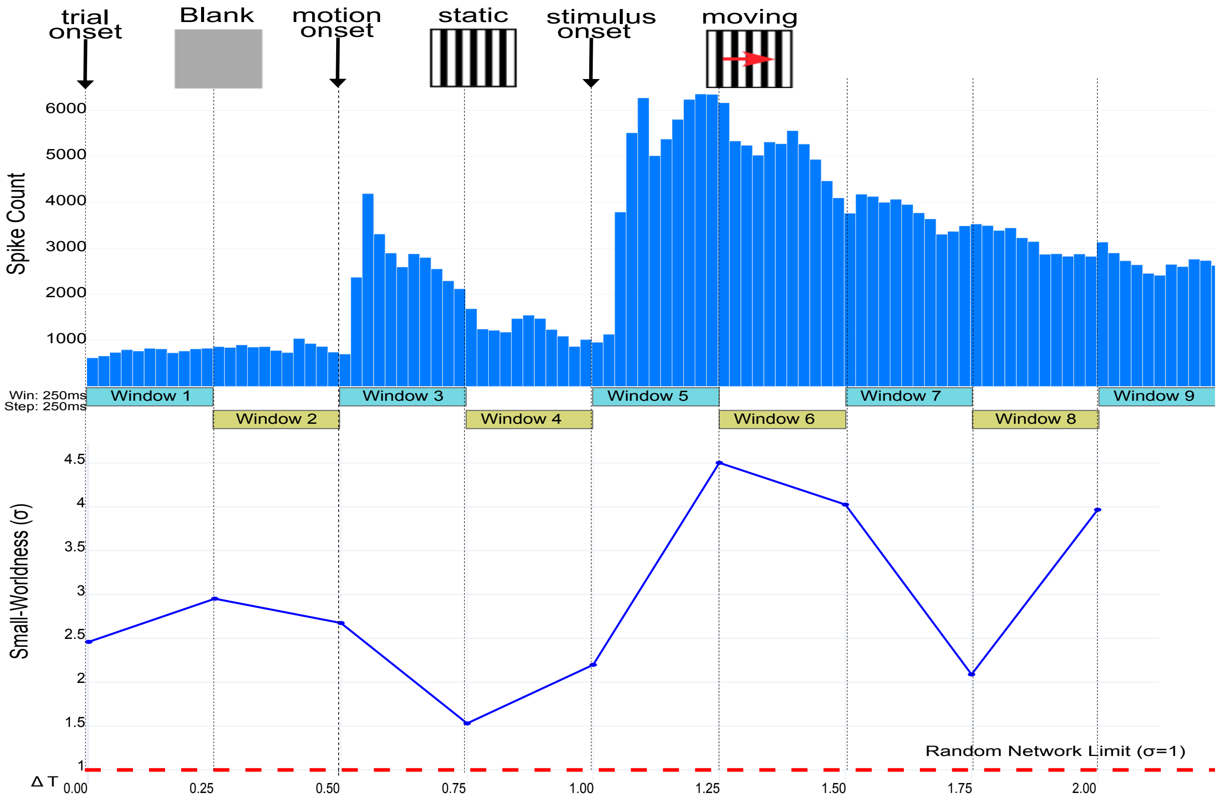

Figure 5 – Temporal alignment of localized neural activity and global network topology. (Top) Aggregate spike count across the recorded channels, highlighting the peak firing rate during the active visual motion epoch. (Bottom) Evolution of the Small-Worldness (σ) index computed over 250 ms sliding windows with a 250 ms step. The network consistently operates above the random network limit ($\sigma = 1$), dynamically optimizing its efficiency in response to sensory ingestion.

While global metrics such as Small-Worldness indicate an overall optimization in network efficiency, the true complexity of the visual dynome is unmasked by observing the specific topological microstates. By extracting the Maximum Spanning Tree (MST) backbones at a proportional density threshold of 23%, we can track the specific causal routing pathways milisecond-by-millisecond.

Figure 6 visualizes these time-resolved structural reconfigurations across the first six epochs. In the initial windows (0.00s–0.50s), the network exhibits a dense, highly clustered state. However, as the processing demands evolve (e.g., Windows 4 and 5), the topology transitions into transient microstates where specific extrastriate and Transition Zone (TZ) nodes become central communication hubs. This rhythmic pulsation—shifting between dense integration and targeted segregation—demonstrates that the A17-TZ-A18 axis does not rely on a rigid communication structure, but rather forms ephemeral pathways to dynamically route visual information.

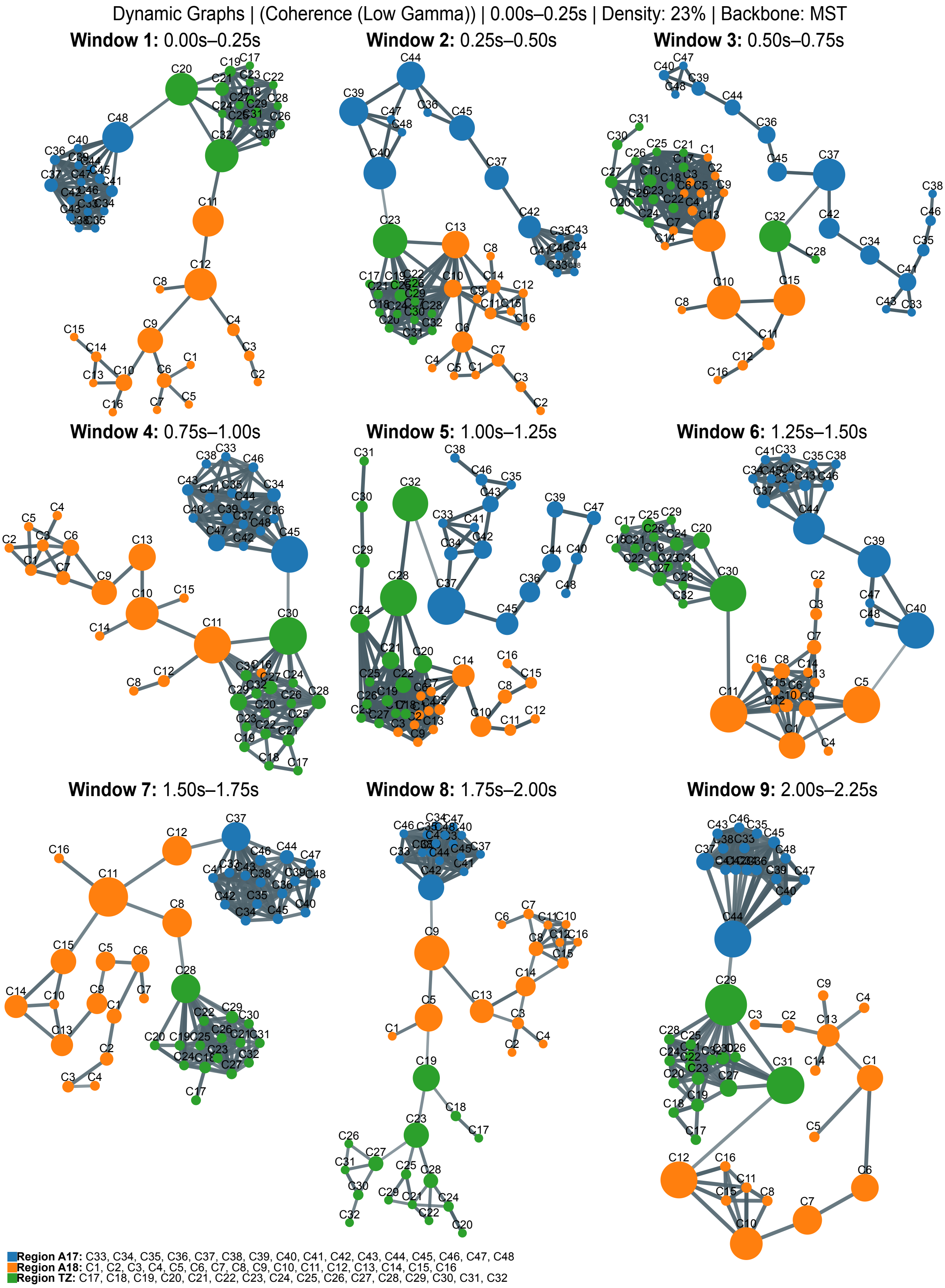

Figure 6 – Time-resolved topological microstates of the visual dynome. The sequence illustrates the dynamic graph coherence in the Low Gamma band across 250 ms sliding windows. To highlight the primary routing architecture and eliminate spurious connections, networks are thresholded at 23% density with a Maximum Spanning Tree (MST) backbone. Node colors correspond to distinct cortical areas: A17 (Blue), Transition Zone (Red), and A18 (Green). The topological volatility confirms the continuous reconfiguration of the cortical hierarchy during active perception.

Although these time-resolved topological microstates visually demonstrate a continuous structural reconfiguration—most notably involving the prominence of the inter-hemispheric border—the macro-graph evolution must be anchored by rigorous nodal statistics to mathematically confirm cortical centrality.

4.4 Statistical Anatomy of the Cortical Hubs

To further evaluate the structural hierarchy of the visual network, we investigated the Node Strength of the cortical areas within the Gamma band. The analysis of the empirical distribution revealed markedly distinct connectivity profiles between primary and higher-order areas.

As evidenced in Figure 7, Area 17 presents a restricted network centrality with low dispersion. The empirical distribution indicates that network integration within the Gamma band is concentrated in the A18-TZ axis, while A17 remains at the structural periphery.

A defining feature of this functional axis is the bimodal signature evident in the raw scatter distributions of both Area 18 and the Transition Zone. The individual data points do not cluster homogeneously around the median; rather, they segregate into two distinct topological regimes: a basal state of low connectivity (clustering near a Node Strength of 5) and discrete surges of substantial systemic integration (reaching values near 30). This pronounced bimodality reveals that the functional architecture is not a stationary state but a highly dynamic mechanism. The network effectively operates in discrete bursts, with the TZ and A18 transiently exhibiting increased hub-like properties to coordinate stimulus processing, before subsequently relaxing into a decoupled state.

Furthermore, the Transition Zone displays a Node Strength visibly superior to that of Area 18. At a proportional density threshold of 15% in the Low Gamma band, the Transition Zone (TZ) exhibited a Node Strength ($M = 9.60$, $SD = 3.50$), compared to A18 ($M = 7.25$, $SD = 3.11$). The statistical analysis revealed a significant difference between the regions (Mann-Whitney $U = 186.0$, $p = 0.030$, effect size $r = 0.453$). This demonstrates that the Transition Zone operates under a connectivity load that statistically surpasses Area 18.

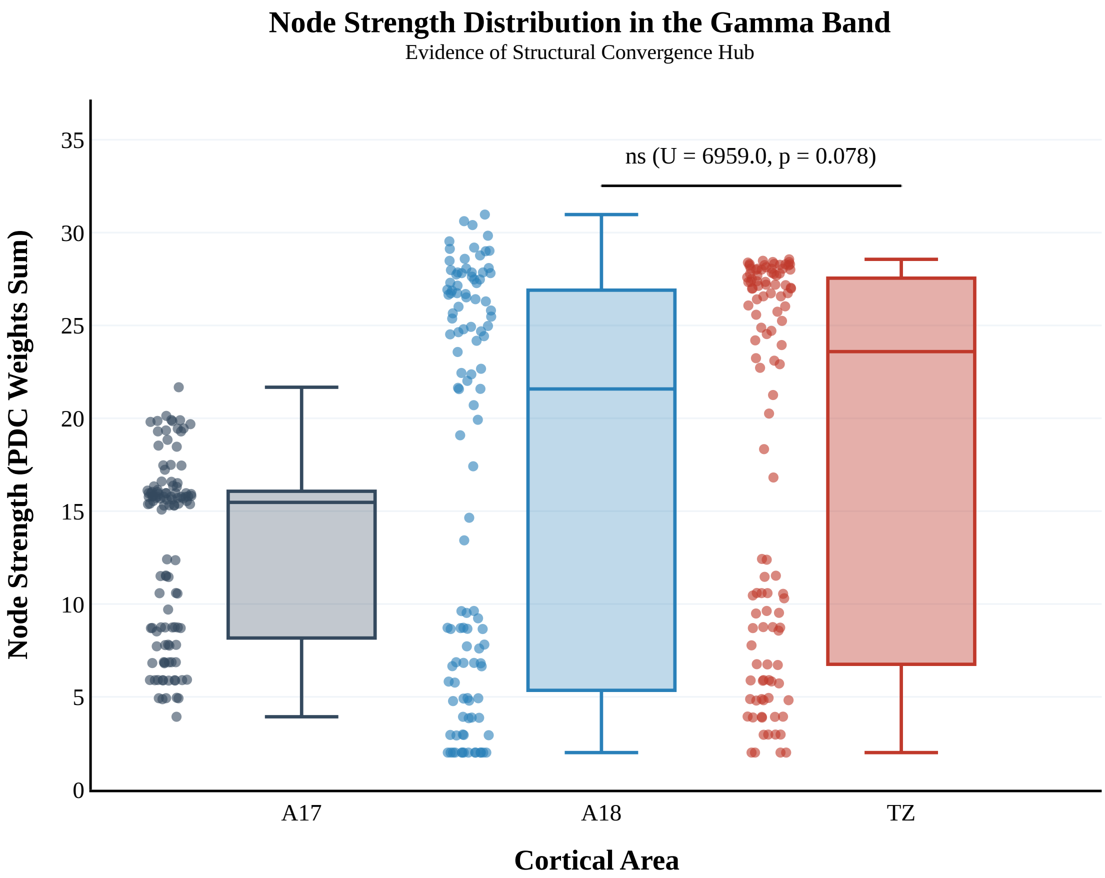

Figure 7 – **Node Strength Distribution in the Gamma Band.** Area 17 displays restricted centrality with low variability. In contrast, both the Transition Zone (TZ) and Area 18 exhibit bimodal distributions with high variability. The Transition Zone operates under a connectivity load tendentially superior to that of Area 18, highlighting its role as a primary interhemispheric convergence hub.

While localized nodal statistics confirm the topological dominance and high processing load of the Transition Zone during active sensory ingestion epochs, structural centrality alone cannot reveal the underlying source of the signaling pathways. To isolate the mechanistic engine driving these reconfigurations, we execute a microsecond-level dissection of the raw directional causal flow.

4.5 Time-Resolved Causal Flux and Network Drivers

A prerequisite for establishing dynamic visual dynome topology is the direct quantification of causal information flow between nodes without the bias of temporal averaging. We utilized the developed NECTA framework to extract the evolution of time-resolved causal coupling using the Partial Directed Coherence (PDC) metric across sensory ingestion epochs.

Figure 8 visualizes these raw directional data. Following stimulus onset (represented by $T = 0$), distinct causal spikes appear that define the functional state. Specifically, Striate cortex (A17) drove the initial response to extrastriate targets (e.g., A17→TZ) with a PDC peaking at 752.4 ms relative to stimulus onset, reaching 0.19 ± 0.13. This data provides the empirical foundation to move beyond descriptive graph metrics, demonstrating that sensory information flow reconfigures rapidly during the active state, transitioning control from striate to extrastriate sources.

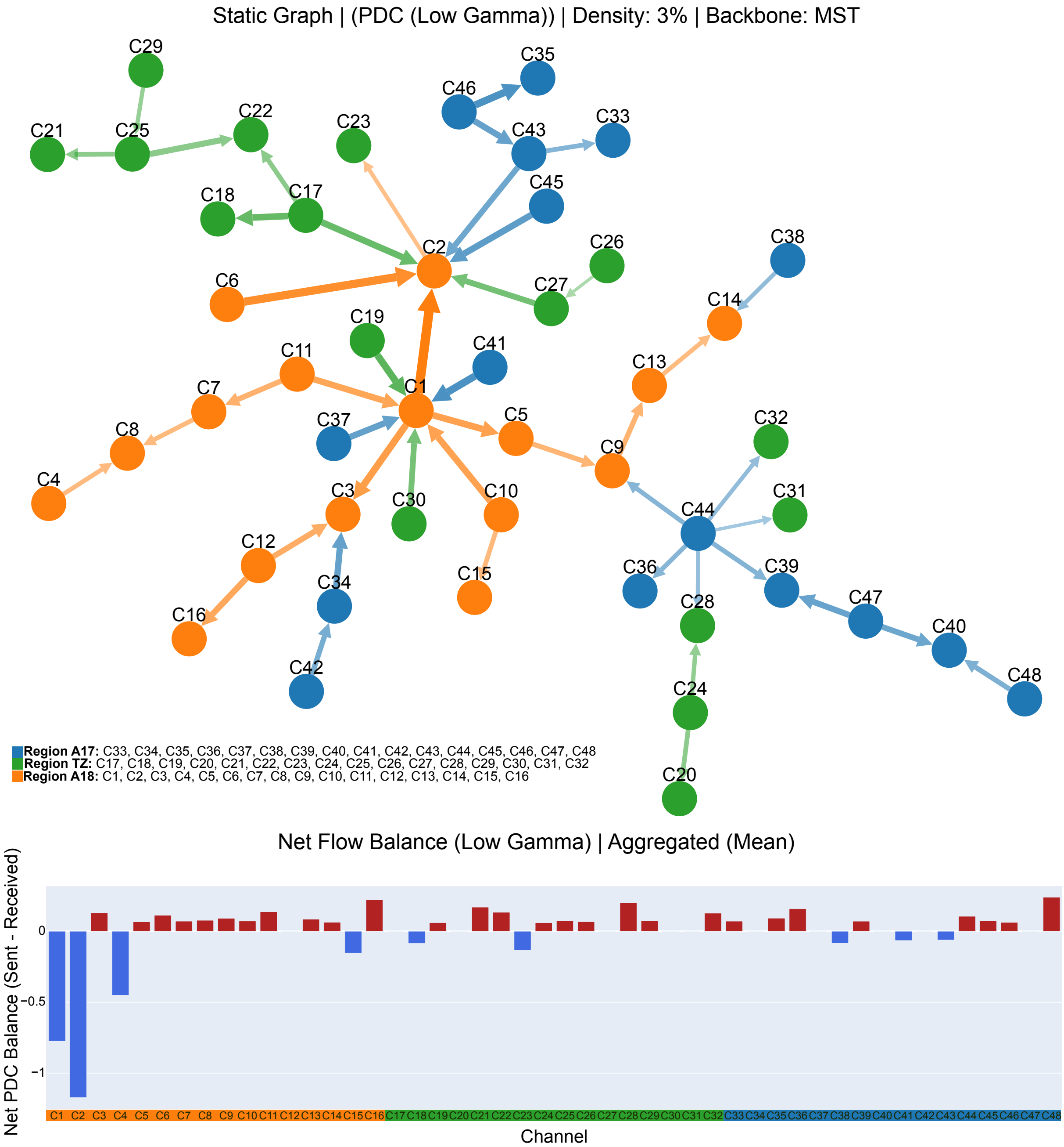

Figure 8 – The figure displays the evolution of Partial Directed Coherence (PDC) over visual ingestion epochs. Colored lines represent distinct directional area connections (e.g., Blue: A17→TZ; Red: A18→TZ). Spikes in the PDC metric indicate active causal information transfer that redefines the visual dynome topology during active sensory states.

To decode the causal relationships within this structural backbone, Partial Directed Coherence (PDC) was applied using the BIC-optimized MVAR models. The static PDC analysis in the Low Gamma band revealed a highly asymmetrical and hierarchical information flow across the visual cortex. By calculating the Net Flow for each region, the analysis indicates that A17 behaves as the dominant persistent Driver, exhibiting a significantly higher Net Flow ($M = 1.21$, $SD = 1.17$) compared to the rest of the network ($M = -0.60$,

$SD = 1.37$; Mann-Whitney $U = 424.0$, $p = 1.245 \times 10^{-4}$, effect size $r = -0.656$). In contrast, A18 behaves as the primary *Receiver* (Sink) (-1.36 net flow). These results suggest a directional pathway extending from A17 through the TZ and terminating in A18 (A17 $\rightarrow$ TZ $\rightarrow$ A18).

The core capability of NECTA lies in capturing the time-varying network dynamics—the transition from static networks to chronological connectomes. Connectivity was re-calculated using a sliding-window approach ($W = 500$ ms, 50% overlap). The Temporal Integration panel overlaid the global population spiking activity (Peri-Stimulus Time Histogram - PSTH) with the topological evolution of the network's *Small-Worldness* (σ).

The dynamic analysis revealed that the network's topological efficiency is not static but responds to visual processing phases. The network achieved its absolute peak in *Small-Worldness* specifically during the moving grating stimulus phase. This transient optimization was driven primarily by a rapid surge in local clustering, indicating that the cortical network reorganizes into a highly efficient state to process dynamic visual inputs.

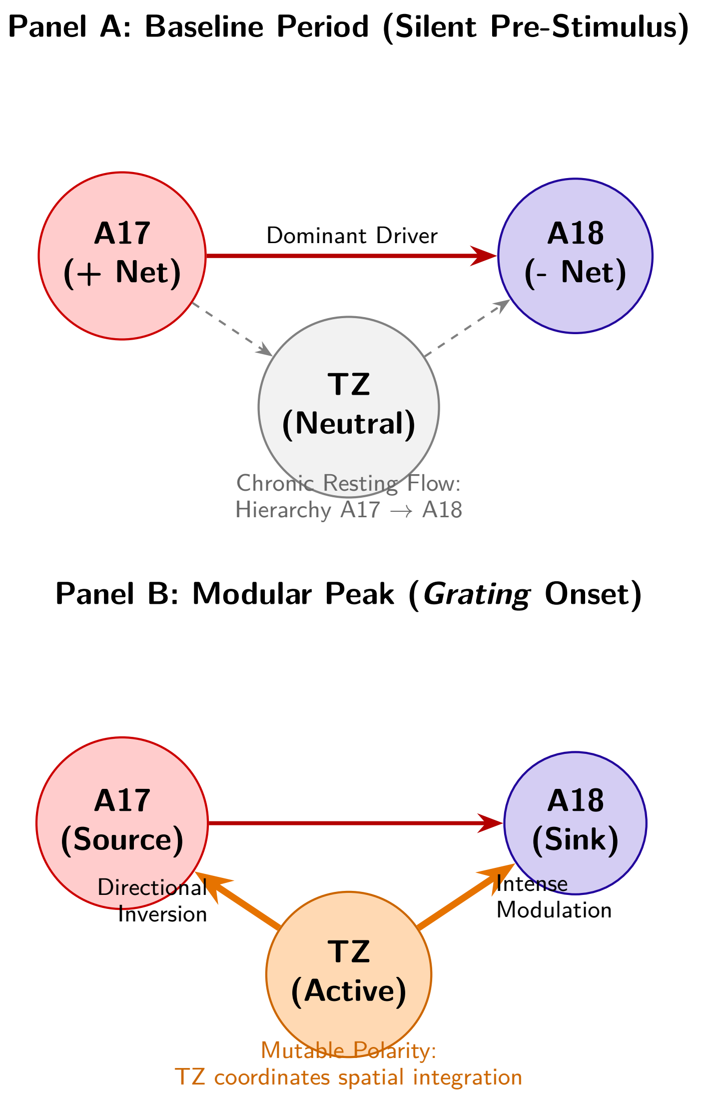

Figure 9 – **Evolutionary Topological Mapping of Information Net-Flow (Low-Gamma Band).** In-degree minus out-degree proportions. **(A)** During the baseline period, the network operates in a predominant predictive visual hierarchy, where Area 17 (A17) is the primary driving pole and Area 18 (A18) the passive sink, with the Transition Zone (TZ) in a neutral state. **(B)** At the precise moment of the stimulus (moving *grating* onset), the topology undergoes active remodeling. The TZ reverses its passive polarity and adopts acute modulation dynamics, drastically altering the *edge weights* (arrow thickness) and suggesting that mesoscopic flexibility reacts transiently to the peripheral stimulus input.

Extracting frame-by-frame snapshots (Connectome Filmstrip) of the dynamic Low Gamma Coherence network illustrated a stable rhythmic organization over time. The hubs located in A17 and the TZ maintained cohesive clustering. Crucially, evaluating the *Net Flow Evolution* (Figure 9) uncovered a complex temporal interplay obscured by static analysis. While A17 consistently acted as a "Volatile Source" and A18 as a "Stable Sink", the Transition Zone (TZ) exhibited unique behavior. The TZ functioned as a *Dynamic Modulator*, actively switching its directional role (from sink to source, or vice versa) precisely at the onset of visual stimulation. This highlights the framework's capacity to dissect the mesoconnectome chronologically, revealing dynamic hierarchical relays that coordinate sensory integration across the interhemispheric boundary.

5 DISCUSSION

5.1 From Static Maps to Dynamic Visual Networks

Historically, the functional organization of the primary visual system has been delineated through the static mapping of receptive fields and anatomical inter-hemispheric tracing (WUNDERLE; ERIKSSON; SCHMIDT, 2013); (SCHMIDT et al., 2023). While these classical methods successfully outline the spatial constraints and functional columns of the cortex, they inherently capture only a time-averaged "structural backbone." Cognitive processing and the integration of moving stimuli, however, depend on millisecond-scale state transitions within neural ensembles—a temporal dimension of rapid network evolution (KOPELL et al., 2014).

Even as our anatomical maps reach unprecedented levels of precision—such as recent efforts updating the classic Binzegger cortical wiring diagram to reveal extensive, previously unaccounted local excitatory synapses terminating in layer 6 (MARTIN; SäGESSER, 2024)—structure alone cannot predict traffic. While anatomy dictates the available pathways, only time-resolved effective connectivity frameworks can reveal how this intricate wiring operates on a millisecond scale.

The application of the NECTA framework to the `c1608a01` dataset provides empirical evidence of this dynamic nature. By utilizing a continuous sliding-window approach, our results suggest that the visual cortex undergoes rapid, stimulus-locked topological reorganizations, reaching transient peaks in *Small-Worldness* precisely during the moving grating stimulus.

This macroscopic network efficiency—an active, responsive property rather than a fixed anatomical trait—is strongly supported by recent cellular and anatomical discoveries. On a microscopic scale, *in vitro* microfluidic studies by (LASSERS; BREWER, 2026) have shown that theta oscillations (4–10 Hz) in isolated axons occur spontaneously and independently of spiking. These oscillations actively coordinate target neuron bursting based on phase and amplitude through a sparse coding architecture, where only a restricted fraction of axons carries high-amplitude signals. This selective, sparse recruitment of communication channels likely mediates the transient optimization detected by NECTA.

Furthermore, (MARTIN; SäGESSER, 2024) identified an extraordinarily extensive horizontal spread within Area 17, with neurons integrating information across distances of up to 11 millimeters (spanning 5–10 degrees of the visual field). This vast lateral arborization provides a plausible anatomical substrate for the network to rapidly bind global spatial information, explaining the acute surges in *Small-Worldness* observed computationally. Their discovery of a novel, direct "shortcut" projection from the layer 3/4 border to layer 6—bypassing the canonical layer 5 relay and likely associated with the fast-conducting, motion-sensitive Y-pathway—further underscores the necessity of high-frequency dynamic analyses. The existence of such fast parallel anatomical routes suggests that visual processing is not strictly serial or slow, fully justifying our focus on the gamma band and sliding-window techniques to capture rapid synchronizations that static tools completely miss.

5.2 Isolating the Feedforward Flow in the Anesthetized State

A critical strength of our biological validation lies in the physiological state of the animal model. The recordings were obtained under general anesthesia and complete paralysis. Historically, anesthesia has often been viewed as a limitation for studying information flow. However, within the modern framework of predictive coding, this state acts as an analytically significant "pharmacological lesion" model. The literature indicates that general anesthesia preferentially suppresses the inhibitory top-down feedback pathways, which are typically mediated by slower alpha and beta oscillations (VEZOLI et al., 2021).

Crucially, while top-down executive modulations are suppressed, the bottom-up feedforward pathways—mediated by high-frequency gamma oscillations—are left to operate autonomously. Therefore, rather than a mere limitation, the anesthetized state elegantly isolated the intrinsic feedforward flow from Area 17 to Area 18. This pharmacological isolation provides a powerful justification for our findings of strong, stimulus-locked directionality restricted to the low-gamma band. It guarantees that the dynamic topological reorganizations and the causal directionality shifts extracted by NECTA are not artifacts of top-down cognitive states, attention, or saccadic eye movements. Instead, the framework successfully unveils the pure, low-level reflexive plasticity inherent to the local synaptic loops and dense transcallosal pathways interlinking A17, A18, and the TZ.

5.3 Mitigating Electrophysiological Confounders

A significant challenge in high-density LFP connectomics is the pervasive artifact of volume conduction, wherein a single electrical source is instantaneously registered by multiple neighboring electrodes (BASTOS; SCHOFFELEN, 2015). If unaccounted for, this phenomenon generates spurious zero-phase lag correlations, heavily biasing the resulting graph topology with false-positive edges.

NECTA addresses this methodological bottleneck architecturally. By employing the Hilbert Transform to extract analytical signals, the framework calculates phase-based indices such as the Weighted Phase Lag Index ($wPLI$) and Imaginary Coherence ($iCoh$), which are mathematically immune to instantaneous spatial spread. This algorithmic refinement facilitates the extraction of dynamic topologies that reflect true underlying physiological synchronization rather than electrical crosstalk. This distinction is critically reinforced by recent experimental evidence isolating single axons in microfluidic tunnels, which suggested that subthreshold voltage oscillations are genuine, regenerative local phenomena of the axon rather than mere volume-conducted epiphenomena of the extracellular matrix (LASSERS; BREWER, 2026). By utilizing *wPLI* and *iCoh*, NECTA's architecture is consistent with this biological reality, mathematically isolating true phase-delayed interactions.

5.4 Gamma Synchronization, Predictive Coding, and Callosal Modulation

Previous work from our group has established the cardinal role of gamma-band synchronization in functional routing and cohesion within the retinogeniculate and cortical pathways of the cat (NEUENSCHWANDER et al., 2023). NECTA extends this foundation by mapping the causal directionality of these high-frequency rhythms.

By applying Partial Directed Coherence (PDC) optimized dynamically via the Bayesian Information Criterion (BIC), we eliminated the bias of static model-order selection (which often leads to overfitting when estimating temporal shifts, unlike AIC). The resulting hierarchy in the Low Gamma band (A17 acting as the persistent Driver, A18 as the Receiver) is consistent with current theories of predictive coding and feedforward information routing (VEZOLI et al., 2021). At the cellular level, this macroscopic directionality is supported by findings that the phase of axonal theta oscillations dictates the firing probability of target neurons, establishing a clear physical causality where subthreshold oscillations 'prime' target neurons for spiking (LASSERS; BREWER, 2026).

A fundamental hurdle when applying multivariate autoregressive (MVAR) based metrics like PDC to continuous electrophysiological recordings is the assumption of signal stationarity, which is heavily violated during sensory processing, particularly by transient evoked potentials. To robustly address this, NECTA tackles non-stationarity through a rigorous multi-stage approach. First, adopting a classical defense well-established in the Granger causality literature, the pipeline performs the subtraction of the ensemble average (the mean evoked potential) from each individual trial. By removing this phase-locked evoked response, the framework successfully isolates purely induced cerebral activity, effectively mitigating the primary source of signal non-stationarity within short sliding windows (e.g., 500 ms). Second, it implements a stationarity filter based on stringent variance thresholds. This filter actively identifies and isolates any remaining epochs of extreme amplitude volatility, ensuring that the sliding windows processed by the MVAR primarily contain stable oscillatory activity. Finally, for windows exhibiting mild non-stationarity breaks, NECTA relies on the dynamic optimization of the Bayesian Information Criterion (BIC). As demonstrated by (FAES; NOLLO, 2010), estimating causality in the frequency domain requires precise control over model complexity to handle instantaneous and lagged interactions without overfitting. The stringent penalty parameter intrinsic to BIC actively limits the autoregressive model order during these state shifts, effectively penalizing spurious complexity. Consequently, this dynamic scaling prevents the PDC estimates from modeling transient artifacts or noise as causal edges, ensuring that the detected causal hierarchies reflect genuine physiological routing.

The validity of utilizing PDC for this structural mapping is strongly supported by recent large-scale computational studies. Nunes et al. (2021) demonstrated *in silico* using spiking neuronal networks that generalized PDC provides a highly reliable estimate of true underlying anatomical structural connectivity (yielding a Pearson correlation of $r = 0.74$). Crucially, their model intrinsically generated LFP signals with prominent spectral peaks in the gamma band, strongly supporting our empirical focus on low-gamma (30–59 Hz) synchronization as a primary channel for causal routing. Furthermore, a persistent challenge in high-density electrophysiology is the “partial observation” problem, where unrecorded brain regions might introduce spurious causalities. (NUNES et al., 2021) indicated that PDC estimates maintain robust correlations with structural connectivity ($r > 0.6$) even when the analysis is restricted to a small spatial cluster. This provides powerful theoretical justification that the causal inferences drawn by NECTA from the restricted A17-TZ-A18 microcircuit remain statistically valid despite the absence of whole-brain recordings. At the cellular level, this macroscopic directionality is supported by findings that the phase of axonal theta oscillations dictates the firing probability of target neurons, establishing a clear physical causality where subthreshold oscillations ‘prime’ target neurons for spiking (LASSERS; BREWER, 2026).

Furthermore, identifying the Transition Zone (TZ) as a "Dynamic Modulator" that reverses its causal flow upon stimulus onset highlights the necessity of time-resolved directional tools. Anatomically, the TZ boundary is densely innervated by mixed intra-hemispheric projections and transcallosal fibers that align spatial orientation maps across the vertical meridian (SCHMIDT et al., 2023). By switching from a passive sink to an active causal source during the moving grating stimulation, the TZ likely orchestrates the interhemispheric temporal unifications required to track sensory borders crossing visual fields. Static connectivity would have merely averaged this reversal into an indiscernible null-flow, masking a critical operational mechanism of visual processing. This substantial functional volatility is theoretically supported by the concept of “activity flow” across neural networks, where the variability of a node’s directed functional connectivity is heavily influenced by its structural centrality (NUNES et al., 2021). Crucially, this dynamic hub-like behavior is grounded in recent retrograde tracing data, which revealed that the border region representing the vertical meridian (the TZ) acts as an epicenter of convergence for both dense long-range ipsilateral projections and substantial contralateral inputs (MARTIN; SäGESSER, 2024). This anatomical reality explains the results observed in our Node Strength analysis, where the spatially restricted TZ statistically surpassed the substantial extrastriate Area 18 in topological centrality. Despite Area 18 being a large region responsible for processing a substantial volume of complex visual features, the TZ's structural wiring as a dense convergence zone causes it to act as an interhemispheric bottleneck under high connectivity load. Thus, the TZ is anatomically predetermined to act as a highly labile dynamic modulator within the A17 → TZ → A18 causal network, gating information flow based on the immediate demands of the moving stimulus.

5.5 Overcoming the Computational Scale and Open Science

Translating the theoretical concepts of dynamic connectomics into practical analytical outputs requires overcoming severe computational barriers. Generating stochastic multivariate null models (e.g., surrogate data phase-randomization and configuration models) across continuous sliding windows triggers a combinatorial explosion that typically crashes single-threaded analytical scripts. By orchestrating these routines through a Dask-distributed High-Performance Computing (HPC) backend and utilizing ‘xarray‘ and ‘Zarr‘ for lazy-loading operations, NECTA effectively eliminates the "Big Data" electrophysiology bottleneck.

Equally vital to this computational leap is the aspiration toward *Open Science* and reproducibility. While systemic barriers remain in widespread adoption, the rigorous parameterization, GitHub/CodeMeta logging, and containerized deployment of NECTA's codebase represent a significant effort to ensure that our analytical workflows are transparent and globally verifiable. By doing so, it provides the ideal computational environment to validate large-scale biological theories, such as integrating subthreshold oscillatory dynamics into next-generation Spiking Neural Networks (SNNs) for more energy-efficient temporal processing (LASSERS; BREWER, 2026). Equally vital to this computational leap is the adherence to *Open Science* and reproducibility. The rigorous parameterization, GitHub/CodeMeta logging, and containerized deployment of NECTA's codebase ensure that our analytical workflows are transparent and globally verifiable. This pipeline aims to lower the barriers to complex graph-theoretical analysis, striving to allow the scientific community to interactively slice and visualize multidimensional connectomes.

6 CONCLUSION AND FUTURE PERSPECTIVES

6.1 Conceptual and Methodological Synthesis

The development of the NECTA (Neural Electrophysiological Connectivity Time-resolved Analysis) framework represents a critical paradigm shift from traditional static graph representations to the continuous exploration of the neural "dynome." Historically, functional connectivity analyses in electrophysiology have collapsed the temporal dimension, effectively masking the highly volatile, millisecond-by-millisecond network state transitions that underlie sensory integration. By establishing a high-performance computational pipeline, this work successfully bridges the gap between raw high-density multi-electrode arrays and rigorous topological inference.

Methodologically, NECTA directly addresses two core bottlenecks in modern neuroinformatics: the Big Data processing limitation and topological false positives. Through zero-phase filtering and the strategic application of robust phase-coupling metrics (such as iCoh and wPLI), the pipeline systematically mitigates the confounding effects of localized spatial crosstalk and volume conduction physical artifacts. Combined with memory-efficient array structures, the framework demonstrates that tracking time-resolved directional workflows is computationally viable without inducing catastrophic Out-Of-Memory (OOM) failures, thereby establishing a reproducible benchmark for large-scale cortical circuit mapping.

6.2 Neurobiological Implications of the Dynome

The empirical application of NECTA to high-density feline electrophysiological recordings yielded substantial insights into the functional architecture of the primary visual cortex. Central to these findings is the unmasking of the localized neurobiological role of the Transition Zone (TZ). Our statistical validation in Chapter 4 demonstrated that under a 15% proportional density threshold in the Low Gamma band (30–59 Hz), the TZ operates with a remarkably high processing load, exhibiting a significantly elevated Node Strength ($M = 9.60$, $SD = 3.50$) compared to the major extrastriate Area 18 ($M = 7.25$, $SD = 3.11$; $p = 0.030$, effect size $r = 0.453$).

Biologically, this strong effect size demonstrates that despite its restricted anatomical boundary compared to the extensive retinotopic expanses of striate and extrastriate cortices, the TZ acts as an ultra-dense inter-hemispheric convergence hub. The high topological variance and causal bimodality captured by NECTA suggest that the TZ serves as a flexible, self-tuning modulator, dynamically routing Source/Sink communication channels depending on the immediate state of visual processing. Crucially, because these non-stationary coupling patterns were recorded under a controlled anesthetized animal model, they reflect fundamental processes of primary intrinsic cortical plasticity, operating entirely independent of top-down conscious attention or cognitive modulations.

6.3 Technical Limitations

An honest assessment of the NECTA framework highlights inherent operational boundaries that shape the interpretation of the dynome. First, a fundamental trade-off between spatial and temporal configurations persists; while multi-electrode electrophysiology yields microsecond-level temporal precision ideal for fast synchronization metrics, its spatial coverage remains anchored to specific, surgically restricted Regions of Interest (ROIs), lacking the holistic whole-brain macro-topology provided by lower-resolution metabolic methods such as fMRI.

Second, the structural topology extracted by the pipeline remains sensitive to operational threshold choices. The enforcement of a specific proportional density limiar (e.g., 15%) and the discretization of continuous time series into brief, arbitrary sliding windows operate as mathematical filters. While these constraints are mathematically required to isolate non-stationary patterns from stochastic noise, they inevitably establish a boundary condition that bounds the overall visibility of the wider connectivity continuum.

6.4 Future Horizons: Towards Algorithmic Invulnerability

To expand upon the architecture established in this thesis, two rigorous operational vectors are planned for subsequent development, targeting robust algorithmic validation and substantial infrastructure scalability.

6.4.1 In Silico Validation Pipeline

Establishing the robust validity of NECTA requires a rigorous computational demonstration using synthetic benchmarks. Future developments will focus on generating highly controlled multivariate synthetic time series (Ground Truth), embedding predefined theoretical frequencies (30–90 Hz) and predominantly orchestrated directional causal paths. By systematically introducing noise degradation (incorporating varying distributions of Pink and White noise), we will evaluate NECTA's resilience to adverse Signal-to-Noise Ratios (SNR). Spatial leakage and physical volume conduction artifacts will be modeled explicitly, allowing the extraction of consistent precision metrics, such as Receiver

Operating Characteristic (ROC) curves and Area Under the Curve (AUC) statistics, to mathematically benchmark true causal detection rates.

6.4.2 HPC Architectural Stress Testing

To decisively tackle the Big Data processing bottleneck inherent to multi-channel electrophysiology, the pipeline must undergo infrastructure-level architectural stress tests. Future work will benchmark standard sequential pandas-based data execution against parallel distributed orchestration managed via Dask. By tracking scalability across a multi-node high-performance computing environment using 1, 2, 4, 8, and 16 concurrent workers, we will map the acceleration profiles, memory footprint limits, and overall latency boundaries of the framework. This benchmarking will ensure NECTA's scalability when executing time-resolved causal algorithms over large-scale, ultra-high-density recording sessions seamlessly.

6.4.3 Predictive Modeling via Graph Neural Networks (GNNs)

While the current NECTA pipeline excels at extracting and quantifying the topological evolution of the dynome, interpreting these extensive sequences of transient graphs presents a substantial analytical challenge. Traditional machine learning approaches often require flattening the network structure, a process that inherently destroys the delicate spatial topology preserved by the framework. To address this, future work will integrate Graph Neural Networks (GNNs), specifically Spatial-Temporal Graph Convolutional Networks (ST-GCNs), directly into the analytical pipeline.

Because GNNs natively ingest time-resolved adjacency matrices as multidimensional tensors, they are uniquely suited to learn latent topological motifs without relying on human-selected macroscopic metrics. We plan to train GNN models to decode distinct sensory states (e.g., visual stimulus orientation or frequency) purely from the microsecond-by-microsecond causal routing patterns. Furthermore, this deep learning integration will allow for predictive modeling of the Transition Zone's (TZ) bimodality, forecasting its rapid functional shifts between Sink and Source states. This will effectively bridge the gap between descriptive graph theory and predictive neural dynamics, transforming NECTA into an artificial intelligence-driven neuroinformatic engine.

6.4.4 Higher-Order Topology: Hypergraphs and Polyadic Interactions

While the traditional graph-theoretical metrics implemented in the current NECTA pipeline provide a robust foundation for analyzing transient connectomics, they are fundamentally constrained by their dyadic nature—modeling interactions strictly as pairwise links between two distinct nodes. However, cortical processing, particularly the sensory integration and inter-hemispheric routing across the A17-TZ-A18 axis, relies heavily on the synchronous firing of multi-neuronal ensembles. This biological reality requires a mathematical framework capable of capturing polyadic (higher-order) interactions.

To address this multidimensional complexity, future iterations of NECTA will explore Hypergraph theory. Unlike standard graphs, a hypergraph allows a single hyperedge to connect any number of nodes simultaneously, effectively modeling true population-level synchrony and network micro-states. Furthermore, by integrating this mathematical approach with deep learning through Hypergraph Neural Networks (HGNNs), the pipeline will be equipped to learn and decode higher-order topological motifs automatically. This advancement will transcend standard pairwise causality, enabling the framework to map complex, non-linear cortical synergies that remain entirely invisible to traditional adjacency matrices.

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